measure_and_integral

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1 Measurable Space

1.1 Measurable Space

In $\mathbb{R}$, we define the volume of an interval to be the length of it. That is, if a closed [1] interval is of the form $I = [a, b]$, then its length is $b - a$. Additionally, we want several properties such as: *additivity*, i.e. if a set is a disjoint union of intervals then the volume of that set is the sum of volumes of the intervals, *complement*, i.e. if a set has volume then the complement also has. It leads to the definitions of σ-algebra and measure.

Definition 1 (σ-algebra) *Let X be an arbitrary set, M is said to be a σ-algebra on X if M consists of subsets of X with properties as follows:*

- $\varnothing, X \in M$
- $U \in M \implies X \setminus U \in M$ *(complement)*
- $U_1, U_2, ... \in M \implies \bigcup_{k\in\mathbb{N}} U_k \in M$ *(countable union[2])*

Definition 2 (Measurable Space) *Let X be an arbitrary set and a σ-algebra M defined on X. A function $\mu : M \to [0, +\infty]$ is called measure if $\mu(\varnothing) = 0$ and given a countable collection of disjoint sets $U_1, U_2, ... \in M$ then $\mu(\bigcup_{k\in\mathbb{N}} U_k) = \sum_{k\in\mathbb{N}} \mu(U_k)$. The triplet (X, M, μ) is called a measurable space. A set $U \in M$ is called a measurable set.*

1.2 Basis of a Measurable Space

Given an arbitrary collection of σ-algebras, the union and intersection are also σ-algebras. That enables us to talk about σ-algebra generated by a basis.

Definition 3 (σ-algebra generated by basis) *Let X be an arbitrary set and B be a collection of subsets of X. Define $\sigma(B)$ to be the smallest σ-algebra containing B, that is the intersection of all σ-algebra containing B. $\sigma(B)$ is also said to be the σ-algebra generated by B.*

Definition 4 (Basis of a σ-algebra) *Let X be an arbitrary set and M be a σ-algebra on X. A subset B of M is said to be a basis of M if $M = \sigma(B)$*

As discussed earlier, to define volume of a set in $\mathbb{R}$, we will construct a σ-algebra consisting all open intervals on $\mathbb{R}$ as follows

Definition 5 (Borel σ-algebra or Borel algebra) *Let $(X, \mathcal{T})$ be a topological space, the Borel σ-algebra on X, denoted by $\mathscr{B}(X)$ is the smallest σ-algebra defined on X containing all open sets.*

The Borel Measurable Space is then defined as

Definition 6 (Borel Measurable Space on $\mathbb{R}^n$) *$(\mathbb{R}^n, \mathscr{B}(\mathbb{R}^n), \mu)$ is the Borel Measurable Space on $\mathbb{R}^n$ if μ is defined by the volume of n-intervals.*

$$\mu\left(\prod_{k=1}^{n}[a_k, b_k]\right) = \prod_{k=1}^{n}(b_k - a_k)$$

[1] similarly for other types of interval
[2] or σ-union

1.3 Subspace of a Measurable Space

Definition 7 (Subspace of a Measurable Space) *Let (X, M, μ) be a measurable space and $E \in M$. Define the subspace (E, M_E, μ_E) by the restriction of M and μ as follows*

- $M_E = \{U \in M : U \subseteq E\}$
- $\mu_E(U) = \mu(U)$ *if* $U \in M_E$

1.4 Complete Measurable Space

I haven't understood enough to give a reason for complete measurable space

Definition 8 (Outer Measure) *Let (X, M, μ) be a measurable space. Define outer measure $\mu_e : \mathcal{P}(X) \to [0, +\infty]$ by the infimum of the measure of all measurable covers*

$$\mu_e(E) = \inf \{\mu(S) : E \subseteq S \in M\}$$

Outer Measure satisfies several properties

- $A \subseteq B \implies |A|_e \leq |B|_e$ (subset)
- $\left|\bigcup_{k\in\mathbb{N}} E_k\right| \leq \sum_{k\in\mathbb{N}} |E_k|$ (σ-subadditivity)

Definition 9 (Complete Measurable Space)

Definition 10 (Completion of a Measurable Space)

Definition 11 (Lebesgue Measurable Space) [3] *Lebesgue Measurable Space on $\mathbb{R}^n$ is defined as the completion of the Borel Measurable Space on $\mathbb{R}^n$*

1.5 Product Measure

Definition 12 (Product Measurable Space) *Let (X, M, μ) and (Y, N, ν) be measurable spaces. Define the product measurable space as $(X \times Y, \sigma(M \times N), \mu \times \nu)$ where the product measure $\mu \times \nu$ is defined as*

$$(\mu \times \nu)(E_X \times E_Y) = \mu(E_X)\nu(E_Y)$$

where $E_X \in M$ and $E_Y \in N$

1.6 Image Measure

2 Measurable Function

Definition 13 (Measurable Function) *Let (X, M, μ) and (Y, N, ν) be measurable spaces. A function $f : X \to Y$ is said to be measurable if*

$$E_Y \in N \implies f^{-1}(E_Y) \in M$$

In this text, we are particularly interested in the case where $Y = [0, +\infty]$ as it is the foundation to define Lebesgue integral. Let's first define a topology on the totally ordered set $[0, +\infty]$

Definition 14 (Order Topology) *Given a totally ordered set $(X, <)$, the order topology $\mathcal{T}$ on X is defined by the basis consists of rays as follows*

$$B = \{\{x :\in X : x < a\} : a \in X\} \cup \{\{x :\in X : b < x\} : b \in X\}$$

The Borel σ-algebra on $[0, +\infty]$ is now defined as follows

Definition 15 (Measurable Function on Half Extended Real Line) *Let $[0, +\infty]$ be equipped with the order topology. Define the Borel measurable space on $[0, +\infty]$ by $([0, +\infty], \mathscr{B}([0, +\infty]), \nu)$ where ν is defined by the length of intervals. Let (X, M, μ) be a measurable space. A function $f : X \to [0, +\infty]$ is said to be measurable if it is a measurable function to the Borel measurable space on $[0, +\infty]$.*

From now on, we will call *Measurable Function on Half Extended Real Line* by *Measurable Function* if there is no confusion. It is trivial to verify the proposition below.

Proposition 1 *Let (X, M, μ) be a measurable space. A function $f : X \to [0, +\infty]$ is measurable if and only if*

$$f^{-1}(a, +\infty] \in M$$

for all $a \in [0, +\infty]$

[3] can also defined as the unique non-trivial measurable space on $\mathbb{R}^n$ that translation invariant

3 Integration

3.1 Simple Function

Definition 16 (Indicator function) *Let $E \subseteq X$, the indicator function of E denoted by $\chi_E : X \to [0, +\infty]$ is defined by*

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{otherwise} \end{cases}$$

Definition 17 (Non-negative simple function) *A function is said to be simple if the set image is finite set. A non-negative simple function $\phi : X \to [0, +\infty]$ can be written as*

$$\phi = \sum_{n=1}^{N} a_n \chi_{E_n}$$

where $N \in \mathbb{N}$, $a_n \in [0, +\infty]$, $\{E_n\}$ are pairwise disjoint, and $X = \bigcup_{n=1}^{N} E_n$

Definition 18 (Integral of a non-negative simple function) *If $\phi = \sum_{n=1}^{N} a_n \chi_{E_n}$ is a non-negative simple function where E_n are measurable w.r.t some measurable space (X, M, μ). We define the integral of ϕ as follows*

$$\int_X \phi = \sum_{n=1}^{M} a_n \mu(E_n)$$

Theorem 1 (Monotone sequence of simple functions) *Given any $f : \mathbb{R}^n \to [0, +\infty]$, there exists a monotone increasing sequence of simple functions $\phi_k : \mathbb{R}^n \to [0, +\infty]$ that converges pointwise to f*

3.2 Integral of a non-negative function

Definition 19 (Integral of a non-negative function) *Let (X, M, μ) be a measurable space and $f : X \to [0, +\infty]$ be measurable. The integral of f is defined as*

$$\int_X f d\mu = \sup \left\{ \int_X g d\mu : g \leq f \right\}$$

Definition 20 (Lebesgue integral of a non-negative function) *Let $(\mathbb{R}^n, M, \mu)$ be the Lebesgue measurable space on $\mathbb{R}^n$ and $f : \mathbb{R}^n \to [0, +\infty]$ be measurable. The Lebesgue integral of f is defined as*

$$\int_{\mathbb{R}^n} f d\mu = \sup \left\{ \int_{\mathbb{R}^n} \phi d\mu : \phi \leq f, \phi \text{ is simple} \right\}$$

Proposition 2 (Area under the graph) *Let $(\mathbb{R}^n, M, \mu)$ be the Lebesgue measurable space on $\mathbb{R}^n$ and $f : \mathbb{R}^n \to [0, +\infty]$ be measurable. Define the set under f as follows*

$$R(f) = \{(x, y) \in \mathbb{R}^n \times \mathbb{R} : 0 \leq y < f(x)\}$$

Then,

$$\mu(R(f)) = \int_{\mathbb{R}^n} f d\mu$$

Definition 21 (Integral of arbitrary function) *Let $(\mathbb{R}^n, M, \mu)$ be the Lebesgue measurable space on $\mathbb{R}^n$ and $f : \mathbb{R}^n \to [-\infty, +\infty]$ be measurable. The Lebesgue integral of f is defined as*

$$\int_{\mathbb{R}^n} f d\mu = \int_{\mathbb{R}^n} f^+ d\mu - \int_{\mathbb{R}^n} f^- d\mu$$

if either $\int_{\mathbb{R}^n} f^+ d\mu$ and $\int_{\mathbb{R}^n} f^- d\mu$ is finite.

Proposition 3 (Linearity of Lebesgue integral)

3.3 Convergence Theorems of a non-negative function

Proposition 4 (Monotone Convergence Theorem)

Proposition 5 (Fatou Lemma)

Theorem 2 (Lebesgue Dominated Convergence Theorem)

3.4 Product Measure Theorems

Proposition 6 (Sectional Property) *Let $(X, M_X, \mu_X), (Y, M_Y, \mu_Y)$ be measurable spaces. Given any $E \in \sigma(M_X \times M_Y)$. For almost every $x \in X$, the set $E_x = \{y \in Y : (x, y) \in E\} \in M_Y$*

Proposition 7 (Sectional Property for Lebesgue Measurable Space) *Let $X = \mathbb{R}^m, Y = \mathbb{R}^n$, E is a Lebesgue measurable set on $X \times Y$. For almost every $x \in X$, the set $E_x = \{y \in Y : (x, y) \in E\}$, namely section, is Lebesgue measurable on Y*

Theorem 3 (Fubini Theorem) *Let $X = \mathbb{R}^m, Y = \mathbb{R}^n$. If $f : X \times Y \to [-\infty, +\infty]$ is Lebesgue integrable, i.e. $f \in L(X \times Y)$, then*

- *For almost every $x \in X$, function $f_x : Y \to [-\infty, +\infty]$ defined by $f_x(y) = f(x, y)$ is Lebesgue integrable, i.e $f_x \in L(Y)$*
- *Function $g(x) : X \to (-\infty, +\infty)$ defined by $g(x) = \int_Y f_x(y)dy$ is Lebesgue integrable, i.e $g \in L(X)$*

And

$$\int_{X \times Y} f = \int_X \left[\int_Y f_x(y)dy \right] dx$$

3.5 Change of Variables

3.6 Lebesgue Differentiation Theorem

3.6.1 Vitali Covering Lemma

Lemma 1 (Zorn Lemma) *Suppose a partially ordered set P has the property that every chain in P has an upper bound in P. Then the set P contains at least one maximal element.*

Lemma 2 (Axiom of Choice) *For any set X of nonempty sets, there exists a choice function f that is defined on X and maps each set of X to an element of that set.*

Lemma 3 (Simple Vitali Covering Lemma) *In a metric space, let F be a collection of balls such that $\sup_{B \in F} r(B) = R < +\infty$, then there is a (countable in a separable metric space) subcollection M of pairwise disjoint balls such that*

$$\bigcup_{B \in F} B \subseteq \bigcup_{B_k \in M} cB_k$$

where cB_k denotes the open ball constructed from B_k by simple enlarging it c times [4].

Let F_n be a collection of balls. Define a partial order on the set of all disjoint collection of balls in F_n as follows: let A, B be any set of disjoint balls in F_n, define $A \leq B$ as $A \subseteq B$.

Given any chain $A_1 \leq A_2 \leq ...$ where $A_i \in F_n$ for all $i \in N$, there is an upperbound $A = \bigcup_{i \in \mathbb{N}} A_i$. By Zorn Lemma, F_n has a maximal subcollection M of disjoint balls.

Let F_1 be defined as follows

$$F_0 = \left\{ B \in F : \frac{R}{2} < r(B) \leq R \right\}$$

Using the procedure above, construct a maximal subcollection M_1 of disjoint of disjoint balls in F_1.

Let F_n be defined as follows

$$F_n = \left\{ B \in F : \frac{R}{2^{n+1}} < r(B) \leq \frac{R}{2^n} \text{ and } B \notin M_1, M_2, ..., M_{n-1} \right\}$$

Given any $B_n \in F$ such that $\frac{R}{2^{n+1}} < r(B_n) \leq \frac{R}{2^n}$, we claim that B_n must intersect one of $M_1, M_2, ..., M_n$. Suppose B does not intersect $M_1, M_2, ..., M_{n-1}$, then by the construction of F_n, $B_n \in F_n$. Furthermore, by maximality of M_n, B either is or intersects one of ball in M_n.

Given any $B_n \in F$ such that $\frac{R}{2^{n+1}} < r(B_n) \leq \frac{R}{2^n}$, let $B_k \in M_k$ $(k \leq n)$ intersects B_n, then $\frac{R}{2^{k+1}} < r(B_k) \leq \frac{R}{2^k}$. Let c_k, c_n be the centers of B_k and B_n, let $a \in B_k \cap B_n$. For any $b \in B_n$,

[4] $\{x \in \mathbb{R}^n : ||x - x_0|| < r\} \mapsto \{x \in \mathbb{R}^n : ||x - x_0|| < cr\}$

$$
\begin{aligned}
d(c_k, b) &\leq d(c_k, a) + d(a, c_n) + d(c_n, b) \\
&\leq r(B_k) + r(B_n) + r(B_n) && (a \in B_k, a \in B_n, b \in B_n) \\
&\leq r(B_k) + \frac{4R}{2^{n+1}} && \left(r(B_n) \leq \frac{R}{2^n}\right) \\
&\leq 5r(B_k) && \left(\frac{R}{2^{n+1}} \leq \frac{R}{2^{k+1}} \leq r(B_k)\right)
\end{aligned}
$$

Hence, any point in any ball in F is in one of the ball $5B_k$ where $B_k \in M_k$, $k \in \mathbb{N}$. Let $M = \bigcup_{k\in\mathbb{N}} M_k$. Given any two balls B_m, B_n with $m < n$, by the construction of F_n, $B_n \in M_n \subseteq F_n$ then B_n does not intersect B_m. Hence, M is a pairwise disjoint subcollection of F such that

$$\bigcup_{B\in F} B \subseteq \bigcup_{B_k\in M} cB_k$$

where $c = 5$. [5].

If the metric space is separable, let $M_Q = \{Q \cap B_k : B_k \in M\}$ where Q is a dense set. Elements of M_Q is non-empty and pairwise disjoint, hence is there a bijection from M to M_Q Furthermore, by Axiom of Choice, there is a bijection from M_Q into a countable set Q. Hence, M is a countable set.

Lemma 4 (Simple Vitali Covering Lemma for Lebesgue measure) *Let $E \subseteq \mathbb{R}^n$ with $\mu_e(E) < +\infty$, let F be a collection of balls covering E, then for any $0 < \epsilon < \mu_e(E)$, there is a countable subcollection M of pairwise disjoint balls such that*

$$\sum_{B_k\in M} \mu(B_k) \geq \beta\mu_e(E)$$

Let $F_n = \{B \in F : r(B) < n\}$, then $(\overline{F_1} \cap E) \subseteq (\overline{F_2} \cap E) \subseteq \ldots$ and $\left(\bigcup_{n\in\mathbb{N}}(\overline{F_n} \cap E)\right) = E$ where $\overline{F_n} = \bigcup_{B\in F_n} B$, hence [6], $\mu_e(E) = \lim_{n\to+\infty} \mu_e(\overline{F_n} \cap E)$. Therefore, given any $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that $\mu_e(E) - \epsilon < \mu_e(\overline{F_n} \cap E)$. By Simple Vitali Covering Lemma on F_n of bounded radius. there is a countable collection M of balls such that

$$\bigcup_{B\in F_n} B \subseteq \bigcup_{B_k\in M} cB_k$$

Hence,

$$
\begin{aligned}
c^n \sum_{B_k\in M} \mu(B_k) &\geq \mu_e\left(\bigcup_{B\in F_n} B\right) \\
&\geq \mu_e(\overline{F_n} \cap E) \\
&\geq \mu_e(E) - \epsilon
\end{aligned}
$$

Write $M = \bigcup_{k=1}^{+\infty} B_k$, then $\lim_{N\to+\infty} \sum_{k=1}^{N} \mu(B_k) = \mu(M) \geq \frac{1}{c^n}(\mu_e(E) - \epsilon)$. Choose appropriate ϵ, there exists $N \in \mathbb{N}$ such that

$$\sum_{k=1}^{N} \mu(B_k) \geq \beta\mu_e(E)$$

where $\beta > 0$

Lemma 5 (Vitali Covering Theorem for Lebesgue measure) *A collection $\mathcal{V}$ of closed balls is said to cover a set E finely if for every $x \in E$ and $\eta > 0$, there is a ball B containing x whose radius is less than η.*

Suppose E is covered finely by a collection $\mathcal{V}$ of balls such that $0 < \mu_e(E) < +\infty$. Then, given any $\epsilon > 0$, there is a countable collection M of disjoint balls such that

$$\mu\left(E \setminus \bigcup_{B_k\in M} B_k\right) = 0 \text{ and } \sum_{B_k\in M} \mu(B_k) < (1+\epsilon)\mu_e(E)$$

[5] we can make c smaller by choosing finer partition of R

[6] By Theorem 3.27, Richard L. Wheeden - Antoni Zygmund - Measure and Integral: An Introduction to Real Analysis - Second Edition

Given any $\epsilon > 0$, choose an open set $G \supseteq E$ such that $\mu_e(E) \leq \mu(G) < \mu_e(E) + \epsilon$.

Let $G_1 = G, E_1 = E$. Let $\mathcal{V}_1$ be constructed by discarding all closed balls in $\mathcal{V}$ that are not contained in G_1. As $\mathcal{V}$ finely covers E_1 and G_1 is open, $\mathcal{V}_1$ still finely covers E_1 [7]. By Simple Vitali Covering Lemma for Lebesgue measure on E with cover $\mathcal{V}_1$, there is a countable subcollection $M_1 \subseteq \mathcal{V}_1$ such that

$$\sum_{B_i \in M_1} \mu(B_i) \geq \beta \mu_e(E)$$

Then,

$$\begin{aligned}
\mu\left(E_1 \setminus \overline{M_1}\right) &\leq \mu\left(G_1 \setminus \overline{M_1}\right) \\
&= \mu(G_1) - \sum_{B_i \in M_1} \mu(B_i) && \text{(balls in } M_1 \text{ are are contained in } G_1\text{)} \\
&< (1-\beta)\mu_e(E_1) + \epsilon \\
&< (1-\beta/2)\mu_e(E_1) && \text{(choose } 0 < \epsilon < \beta\mu_e(E_1)/2\text{)}
\end{aligned}$$

Let $M^{(k-1)} = \bigcup_{i=1}^{k-1} \overline{M_i}$ where $\overline{M_i} = \bigcup_{B \in M_i} B$. Similarly, Let $G_k = G \setminus M^{(k-1)}, E_k = E \setminus M^{(k-1)}$ be the remaining set after choosing $k-1$ finite subcollection M_i. Construct V_k by discarding all closed balls in $\mathcal{V}$ that are not contained in G_k. As $\mathcal{V}$ finely covers E_k and G_k is open, $\mathcal{V}_k$ still finely covers E_k. Simple Vitali Covering Lemma for Lebesgue measure on $E \setminus M^{(k-1)}$ with cover $\mathcal{V}_1$, there is a countable subcollection $M_k \subseteq \mathcal{V}_k$ such that

$$\sum_{B_i \in M_k} \mu(B_i) \geq \beta \mu_e(E_k)$$

Then, note that $\mu(G_k) = \mu(G) - \mu(M^{(k-1)}) < [\mu_e(E) - \mu(M^{(k-1)})] + \epsilon \leq \mu_e(E_k) + \epsilon$

$$\begin{aligned}
\mu_e(E_{k+1}) = \mu_e\left(E_k \setminus \overline{M_k}\right) &\leq \mu\left(G_k \setminus \overline{M_k}\right) \\
&= \mu(G_k) - \sum_{B_i \in M_k} \mu(B_i) && \text{(balls in } M_k \text{ are are contained in } G_k\text{)} \\
&< (1-\beta/2)\mu_e(E_1)
\end{aligned}$$

Therefore,

$$\mu(E_k) < (1-\beta/2)^{k-1}\mu(E)$$

Let $M = \bigcup_{i \in \mathbb{N}} M_i$, todo

3.6.2 Lebesgue Differentiation Theorem

Lemma 6 (Hardy-Littlewood Maximal Function) *Given a Lebesgue measurable function $f : \mathbb{R}^n \to \mathbb{R}$, Hardy-Littlewood maximal function defined on ball is as follows*

$$f^*(x) = \sup_{r>0} \frac{1}{\mu(\mathcal{B}_r(x))} \int_{\mathcal{B}_r(x)} |f(y)| dy$$

Some properties of Hardy-Littlewood maximal function:

1. *f^* is lower semicontinuous, i.e. $E_t = \{x \in \mathbb{R}^n : f^*(x) > t\}$ is open for all $\alpha \in \mathbb{R}$*
2. *$\mu(E_t) \leq \frac{c}{t} \int_{\mathbb{R}^n} |f(y)| dy$ where $c \in \mathbb{R}$*

(Part 1) Given any $\alpha > 0$, let $x_1 \in E_t = \{x \in \mathbb{R}^n : f^*(x) > t\}$ that is, there exists $r > 0$ such that

$$\frac{1}{\mu(\mathcal{B}_r(x_1))} \int_{\mathcal{B}_r(x_1)} |f(y)| dy > \alpha t$$

where $\alpha > 1$. We will prove that there exists an open ball $\mathcal{B}_\epsilon(x_1)$ such that for every $x_2 \in \mathcal{B}_\epsilon(x_1)$, $f^*(x_2) > t$, that implies the set E_t is open.

Let $\epsilon > 0$, for any $x_2 \in \mathcal{B}_\epsilon(x_1)$, then $\mathcal{B}_r(x_1) \subseteq \mathcal{B}_{r+\epsilon}(x_2)$. Therefore

$$\begin{aligned}
t &< \frac{1}{\alpha} \frac{1}{\mu(\mathcal{B}_r(x_1))} \int_{\mathcal{B}_r(x_1)} |f(y)| dy \\
&\leq \frac{1}{\alpha} \frac{\mu(\mathcal{B}_{r+\epsilon}(x_2))}{\mu(\mathcal{B}_r(x_1))} \frac{1}{\mu(\mathcal{B}_{r+\epsilon}(x_2))} \int_{\mathcal{B}_{r+\epsilon}(x_2)} |f(y)| dy
\end{aligned}$$

[7] A similar argument: for each $x \in G_1 \cap E$, choose one a sequence of decreasing closed balls containing x that is contained in $G_1 \cap E$

Choose ϵ small enough such that $\frac{\mu(\mathcal{B}_{r+\epsilon}(x_2))}{\mu(\mathcal{B}_r(x_1))} < \alpha$, then

$$\frac{1}{\mu(\mathcal{B}_{r+\epsilon}(x_2))}\int_{\mathcal{B}_{r+\epsilon}(x_2)} |f(y)|dy > t$$

That is, $f^*(x_2) > t$. Hence, there exists $\epsilon > 0$ such that for all $x_2 \in \mathcal{B}_\epsilon(x_1)$, $f^*(x_2) > t$. E_t is open for any t.

(Part 2) For each $x \in E_t$, let B_x be a ball centered at x such that $\frac{1}{\mu(B_x)}\int_{B_x} |f(y)|dy > t$. The collection $\{B_x : x \in E_t\}$ covered E_t, by Vitali Covering Lemma, given any $\epsilon > 0$, there is a finite subcollection M_ϵ of pairwise disjoint B_k such that

$$\sum_{B_k \in M_\epsilon} \mu(B_k) \geq \frac{1}{5}\mu(E_t) - \epsilon$$

Therefore,

$$\begin{aligned}
\mu(E_t) &\leq 5 \sum_{B_k \in M_\epsilon} \mu(B_k) + 5\epsilon \\
&< 5 \sum_{B_k \in M_\epsilon} \left(\frac{1}{t}\int_{B_k} |f(y)|dy\right) + 5\epsilon && \text{(by construction of } B_k) \\
&\leq \frac{5}{t}\int_{\mathbb{R}^n} |f(y)|dy + 5\epsilon && (B_k \text{ pairwise disjoint})
\end{aligned}$$

The statement is true for all $\epsilon > 0$, hence

$$\mu(E_t) \leq \frac{5}{t}\int_{\mathbb{R}^n} |f(y)|dy$$

Lemma 7 (Density of Continuous Function) *Given $f \in L(\mathbb{R}^n)$ and any $\epsilon > 0$, there exists a continous function g with compact support* [8] *such that*

$$\int_{\mathbb{R}^n} |f(x) - g(x)|dx < \epsilon$$

Theorem 4 (Lebesgue Differentiation Theorem) *Given $f \in L(\mathbb{R}^n)$, then for almost every $x \in \mathbb{R}^n$*

$$\limsup_{B \searrow x} \left|\frac{F(B)}{|B|} - f(x)\right| = \limsup_{B \searrow x} \frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| = 0$$

where $B \searrow x$ denotes $B = \mathcal{B}_r(x)$ and $r \to 0^+$, $F(B) = \int_B f(y)dyt$

Let D_t be defined as

$$D_t = \left\{x \in \mathbb{R}^n : \limsup_{B \searrow x} \frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| > t\right\}$$

We will prove that $\mu_e(D_t) = 0$ for all $t > 0$, that will conclude the Lebesgue Differentiation Theorem.

Let g be a continous function, and any ball B centers at x, then

$$\begin{aligned}
\left|\int_B [f(y) - f(x)]dy\right| &\leq \int_B |f(y) - f(x)|dy \\
&\leq \int_B |f(y) - g(y)|dy + \int_B |g(y) - g(x)|dy + \mu(B)|g(x) - f(x)|
\end{aligned}$$

Hence,

$$\begin{aligned}
\limsup_{B \searrow x} \frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| &\leq \limsup_{B \searrow x} \frac{1}{\mu(B)}\left(\int_B |f(y) - g(y)|dy + \int_B |g(y) - g(x)|dy + \mu(B)|g(x) - f(x)|\right) \\
&\leq \limsup_{B \searrow x} \frac{1}{\mu(B)}\int_B |f(y) - g(y)|dy + \limsup_{B \searrow x} \frac{1}{\mu(B)}\int_B |g(y) - g(x)|dy + \limsup_{B \searrow x} |g(x) - f(x)|
\end{aligned}$$

[8] support of $f : X \to \mathbb{R}$ is $\{x \in X : f(x) \neq 0\}$

As g is continuous, $\limsup_{B\searrow x} \frac{1}{\mu(B)} \int_B |g(y) - g(x)|dy = 0$, then

$$\begin{aligned}
\limsup_{B\searrow x} \frac{1}{\mu(B)} \left| \int_B [f(y) - f(x)]dy \right| &\leq \limsup_{B\searrow x} \frac{1}{\mu(B)} \int_B |f(y) - g(y)|dy + |g(x) - f(x)| \\
&\leq \sup_B \frac{1}{\mu(B)} \int_B |f(y) - g(y)|dy + |g(x) - f(x)| \\
&= (f-g)^*(x) + |g(x) - f(x)|
\end{aligned}$$

Therefore,

$$D_t \leq \left\{ x \in \mathbb{R}^n : (f-g)^*(x) > \frac{t}{2} \right\} \cup \left\{ x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2} \right\}$$

As the two sets are measurable

$$\mu_e(D_t) \leq \mu\left(\left\{ x \in \mathbb{R}^n : (f-g)^*(x) > \frac{t}{2} \right\} \right) + \mu\left(\left\{ x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2} \right\} \right)$$

By property of Hardy-Littlewood maximal function,

$$\mu\left(\left\{ x \in \mathbb{R}^n : (f-g)^*(x) > \frac{t}{2} \right\} \right) \leq \frac{2c}{t} \int_{\mathbb{R}^n} |f(y) - g(y)|dy$$

By Chebyshev inequality,

$$\mu\left(\left\{ x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2} \right\} \right) \leq \frac{2}{t} \int_{\mathbb{R}^n} |f(y) - g(y)|dy$$

Therefore,

$$\mu_e(D_t) \leq \frac{2(c+1)}{t} \int_{\mathbb{R}^n} |f(y) - g(y)|dy$$

Given any $t > 0$ and $\epsilon > 0$, we can choose g close enough to f such that $\int_{\mathbb{R}^n} |f(y) - g(y)|dy < \frac{t\epsilon}{2(c+1)}$, then $\mu_e(D_t) < \epsilon$. Hence $\mu_e(D_t) = 0$ for all $t > 0$. Then

$$D = \left\{ x \in \mathbb{R}^n : \limsup_{B\searrow x} \frac{1}{\mu(B)} \left| \int_B [f(y) - f(x)]dy \right| > 0 \right\} = \bigcup_{k\in\mathbb{N}} D_{1/k}$$

Then $\mu_e(D) = 0$, that is $\limsup_{B\searrow x} \frac{1}{\mu(B)} \left| \int_B [f(y) - f(x)]dy \right| = 0$ for almost every $x \in \mathbb{R}^n$